FIT5145 - Assignment 3: Business & Data Case Study - Report

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GENERATIVE AI DECLARATION:

In this assignment, I used ChatGPT, Gemini Chatbot to brainstorm, improve writing and clarity, and explain the methodology, concepts.

Have you selected a topic for Assignment 3 that is different from the one that you used for Assignment 1 (i.e., have you rewritten the first three sections of the report)?

No I did not change

Feedback from Assignment 1

I have received the feedback from Kai Dang:

- "*The role of data in the business model (sources, handling, integration) could be articulated more explicitly"*
- *"The multi-disease + multi-factor idea is promising, but the technical novelty is described at a high level and lacks concrete methodological detail."*

From the first feedback, I aim to clarify on the sources of data, how data should be handled, and how different various data are combined into the new suggested model. I modified the business model 3.1 section.

From the second feedback, I aim to modify the novelty 2.3 section, especially the technical aspect**.** I will include the actual technical methods that will be used in this new model and the difference of how data pipelines are built differently than existing models.

Table of Content

1. Introduction

Nowadays, local governments, data scientists, and healthcare professionals create and use big data models to predict the development of mosquito-borne infections, spreading trends, and their possibility of returning to the community **[1]**. However, many well-designed models are developed to predict and study only a single disease, which significantly limits detection and delays local response times when a new mosquito-borne disease outbreak occurs within the region. Mosquito populations have the ability to reproduce rapidly in large numbers within less resourceful environments like stagnant water and urban sewer systems. The ability to carry dangerous human diseases makes mosquitoes highly dangerous. In 2024, there were more than 282 million malaria cases with 610,000 deaths **[2]**. This shows that the spread of mosquito-borne infections is fast and hard to control. In addition, it also demonstrates that existing models are not effective at predicting and stopping the spread of mosquito-borne diseases.

This project aims to provide data to introduce an innovative methodology to improve the forecasting of mosquito-borne diseases. The proposed model is capable of detecting multiple diseases simultaneously. It also determines which areas are most vulnerable to certain infections, and the ecological factors that drive disease spread. This project promises to solve the current problem.

2. Related Work

2.1. Existing projects and works

Three existing works that identify mosquito transmission are considered. First, the dengue forecasting Model Satellite-based System (D-MOSS) is the system developed by the UK company - HR Wallingford, and is used in countries like Vietnam, Malaysia, Sri Lanka **[3]**. This system can forecast the outbreak of the dengue fever 6 months in advance by collecting the outcome of the water availability forecasts model, local province's environmental data, and historical dengue cases **[3]**.

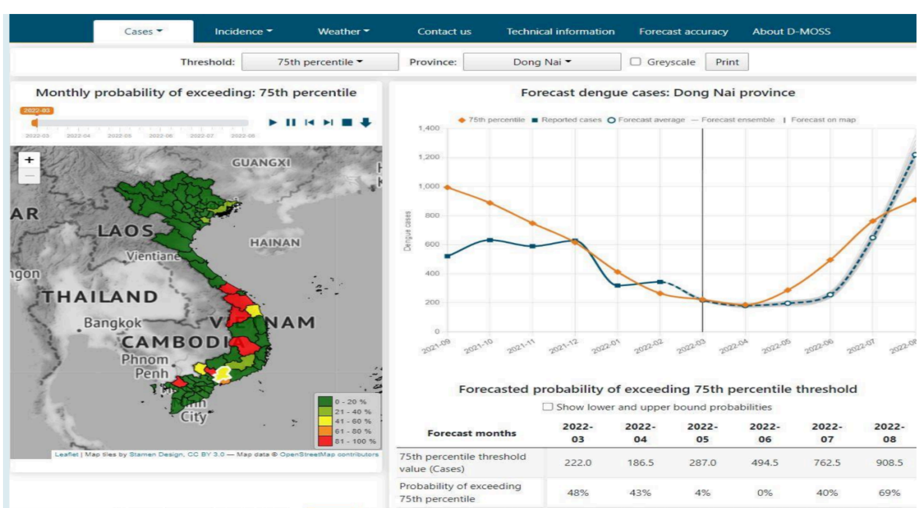


Image 1: The interface of the D-MOSS model **[3]**

Second, Victoria's mosquito surveillance report in Australia is used to notify the public of existing disease in local areas. It traps and extracts diseases carried by local mosquitoes, and collects disease statistics from local hospitals **[4]**.

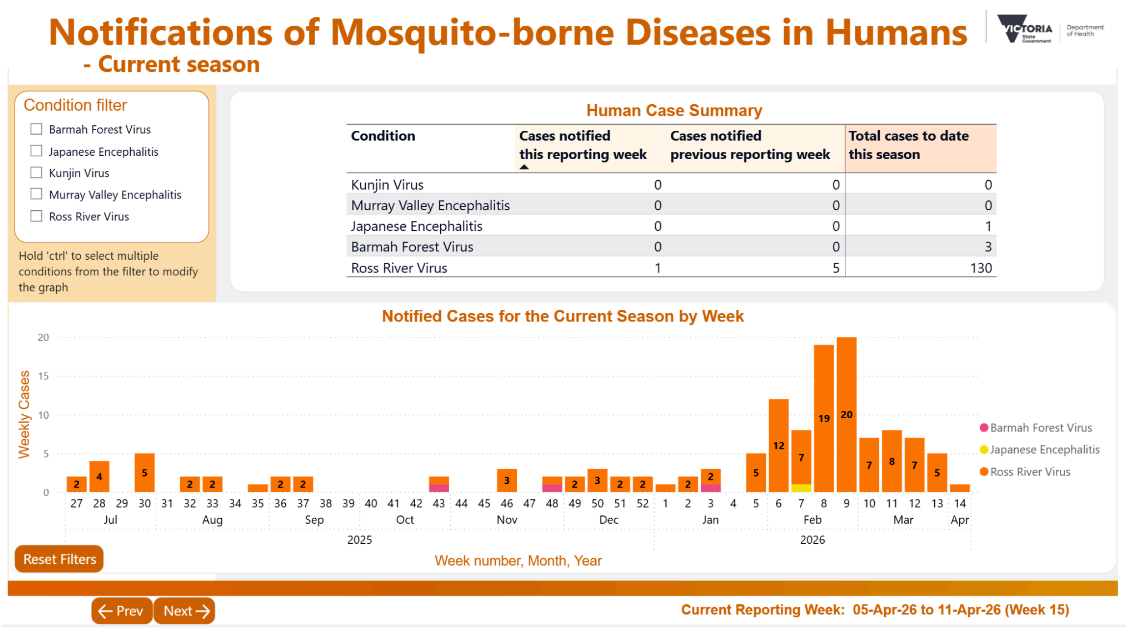


Image 2: The interface of the Victoria's mosquito surveillance report **[4]**

Finally, the Malaria early warning system (MEWS) is used in Africa and India to predict the probability of malaria happening in the next several months **[5]**. It primarily collects data such as temperature, rainfall rate, and vegetation indices which are the main factors of malaria **[5]**.

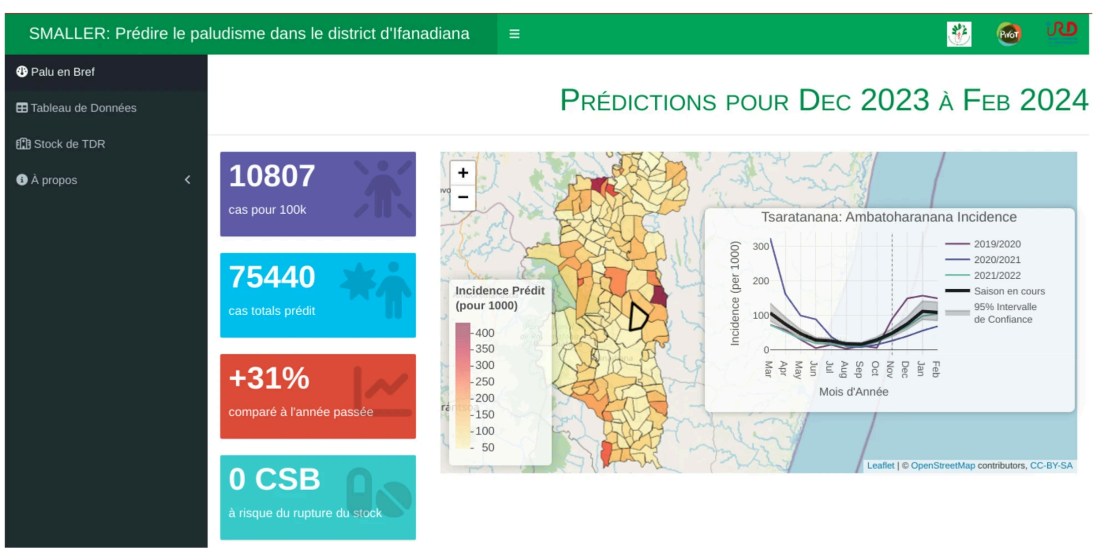


Image 3: The interface of the MEWS model **[5]**

2.2. Gap identification

Although each model demonstrates a great ability to provide insights and predictions, these models do not consider the vulnerability of the community to many diseases. For example, these models do not demonstrate the proportion of the population that is likely to be infected, such as children under 5, pregnant women, and people with compromised immune systems. The three models do not concentrate on providing insights into the community's response to preventing the disease. There is no data related to local vaccination rates, or hospitals and clinics that could respond immediately to mosquito-borne illnesses. Not only that, but there is also little information about the local government's actions to control the mosquito population and eradicate disease-carrying mosquitoes. Models like D-MOSS and MEWS are designed to identify specific diseases, which limits the ability to identify and prevent unpredicted diseases. Victoria's mosquito surveillance report does not provide real-time information, which makes it difficult to supply data for an immediate response.

2.3. Why is the project novel

Many mosquito-borne diseases will appear more widely due to the rise of global temperatures and the development of globalisation, which makes existing models less effective. This project is novel because it implements a unified Multi-Target Machine Learning architecture instead of single-target regression model. This allows the system to simultaneously detect multiple diseases in a single run. Furthermore, to handle the multi-factor aspect, the methodology will incorporate advanced feature engineering, combining macro-level environmental variables (temperature, humidity) with micro-level social vulnerability indices (vaccination rates, age demographics, hospital beds) into a unified predictive algorithm.

3. Business Model

3.1. Business area

This project sits within the data science, epidemiology, and public health informatics domains to deliver timely disease insights to local authorities and the public. For data sources, the model will collect the climate metrics, electronic health informatics, and mosquito data from the meteorological stations, local hospitals and laboratories respectively. For data handling, raw data will be stored in a cloud-based data lake to safely accommodate the high volume and variety. For data integration, Extract, Transform, Load (ETL) processes will be used to merge unstructured geospatial data with structured demographic ones, powering the real-time predictive dashboard.

3.2. Benefits and values

This project is expected to bring various medical, financial, and environmental benefits to the local community. For medical benefits, it will provide information to healthcare professionals like doctors, nurses to have better treatment strategies and decisions to lower the death rate. It will help financially and strategically the local government in allocating resources such as personnel, medical facilities, and budget to economically and early detect and control epidemics. It will also provide evidence to boost community health protection activities, such as cleaning residential areas and raising funds for mosquito eradication. Finally, it helps balance the ecosystem by controlling the mosquito population and nests, which prevents the dominance and imbalance of the natural food chain.

3.3. Stakeholders

This project will concentrate on providing deeper insights for local residents, government, healthcare professionals, and environmentalists. For local residents, the data will provide them with awareness about the risks in their surrounding areas and help them take action to prevent the spread, such as keeping their houses clean and wearing protection for themselves and their children to avoid mosquito bites. For the local government, the data will provide insights to prevent disease outbreaks in their early stages. During epidemics, they can better allocate each area's resources, such as healthcare professionals, medical supplies, and facilities, to control the disease. For hospitals and healthcare professionals, such as doctors and nurses, this project will provide them with knowledge of existing disease causes in the area, allowing them to prepare specific diagnostics and specialised wards before an outbreak hits. For environmentalists, they will have the chance to keep track of mosquito habitats over time and ensure the balance of the ecosystem.

4. Characterising and Analysing Data

4.1. NIST framework

Data Source: Meteorological stations and satellite streams provide regional environmental data, including temperature, humidity, and stagnant water coverage. Public health informatics from hospitals and medical facilities supply clinical metrics such as vaccination rates, historical case counts, and disease-specific bed capacities. Similar to Victoria's health reports, mosquito population data and viral load metrics are sourced from local laboratories.

Volume: The core dataset contains 15 attributes and expands monthly across over 195 countries, encompassing multiple regional provinces and various newly discovered diseases. Additionally, the continuous influx of daily environmental data and monthly textual reports from laboratories and hospitals will scale the total volume to several gigabytes annually.

Velocity: Data from meteorological stations, satellites, and local laboratories exhibit high velocity, requiring daily or weekly ingestion for continuous analysis. Conversely, clinical data from hospitals and medical facilities operate at a lower velocity, with updates occurring on a monthly or quarterly basis.

Variety: The framework must process diverse data formats, including structured tabular data (e.g., funding and bed counts), semi-structured data (e.g., satellite raster grids), and unstructured data (e.g., textual laboratory and hospital reports).

Veracity: The project must mitigate significant data quality risks, including missing environmental data caused by outdoor sensor dropouts, inaccurate mosquito counts due to improper trap setups, and erroneous hospital bed records stemming from human input errors.

Software: Apache Storm will process real-time daily weather updates, while Apache Hive organizes massive tabular datasets for SQL querying. Apache Spark will power the machine learning operations, and the Watson REST API will parse unstructured textual medical reports. AWS will serve as the central cloud storage environment. Locally, RStudio and Python will handle data wrangling, exploratory analysis, and architectural model design.

Analytics: The system functions as a descriptive tool to visualize historical epidemiological trends and employs advanced predictive modeling to accurately forecast potential disease outbreaks.

Processing: The pipeline must continuously ingest high-frequency daily weather and satellite streams while concurrently batch-processing large monthly clinical reports and periodically retraining the machine learning models.

Capabilities: The platform will aggregate data from multiple sources, extract critical features from both structured reports and unstructured geographic streams, and generate real-time health risk alerts.

Security/Privacy: Because the system processes sensitive clinical and healthcare metrics, it requires strict adherence to data privacy protocols and the preservation of patient anonymity.

Lifecycle: The data lifecycle demands continuous maintenance; historical data must be securely archived, and any structural shifts in incoming data formats must be systematically managed.

Other: The framework must dynamically account for varying clinical mosquito reporting standards, diagnostic thresholds, and data formats across different international jurisdictions.

4.2. Data analysis and statistical methods

Multiple Linear Regression (MLR): This model establishes the relationship between multiple independent variables (susceptibility, mosquito density) and dependent variables (case counts) as an example model: *case_count = beta0 + beta1(Mosquito_Density) + beta2(Susceptibility) - beta3(Vaccination_rate)*. It provides high mathematical interpretability by generating specific regression coefficients that quantify the exact directional impact of each predictor.

Long Short-Term Memory (LSTM): This architecture is designed to handle sequential time-series data and retain long-term historical information. In disease prediction, environmental parameters exhibit temporal lags; for instance, variations in humidity and temperature alter mosquito density weeks before translating into higher case counts. LSTMs utilize internal gating mechanisms to effectively store this historical information and learn these environmental behaviors, enabling the model to provide early warnings for upcoming changes in mosquito quantities.

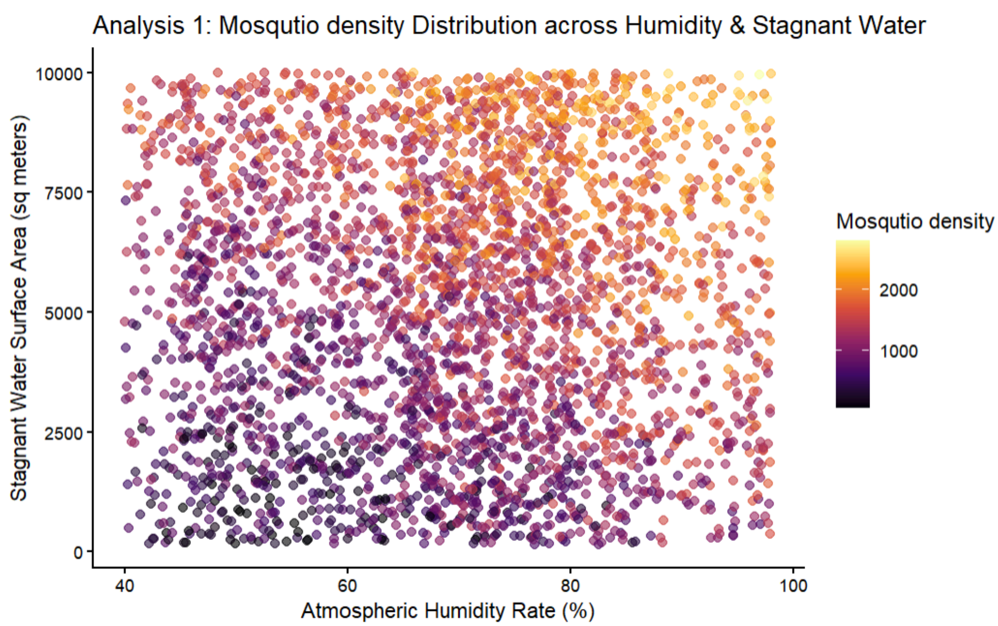
Multilayer Perceptron (MLP): This foundational deep learning network consists of fully connected layers of artificial neurons that pass information forward to map inputs to targets. The MLP is utilized to capture non-linear interactions between multi-source public health datasets. For example, it models how socio-demographic factors and active mosquito densities interact to trigger outbreaks, uncovering multi-dimensional risk patterns that standard linear methods completely miss.

K-Means Clustering: This unsupervised data analysis method groups multi-dimensional data points into distinct by minimizing the distance between data points and their cluster centroids. K-Means automatically groups health regions based on their specific infrastructure deficits, funding gaps, and environmental risks. This grouping reveals shared vulnerabilities and organizes regional health systems into clear priority tiers, allowing governments to target support effectively without needing pre-labeled historical data.

4.3. Demonstration

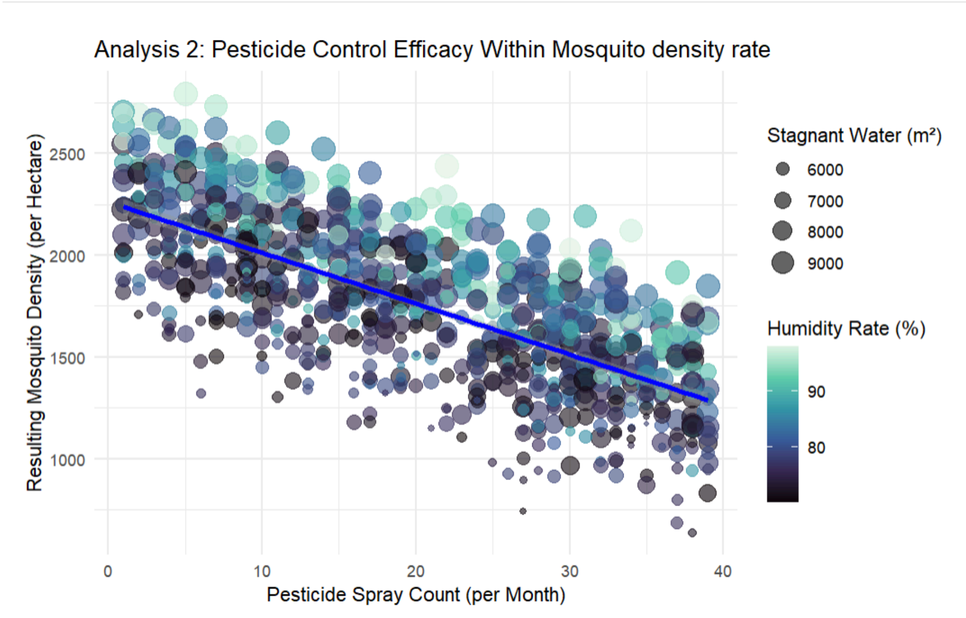
The dataset of the below analysis can be found [here](). Overall, this analysis demonstrates the relationship between the environmental data, pesticide control, population susceptibility, vaccination rate, hospital bed with mosquito density and cases counts perspectively. The data wrangling process focuses on cleaning corrupted or extreme observations within the case counts attribute. The primary insights are detailed below:

A. **Environmental factors**



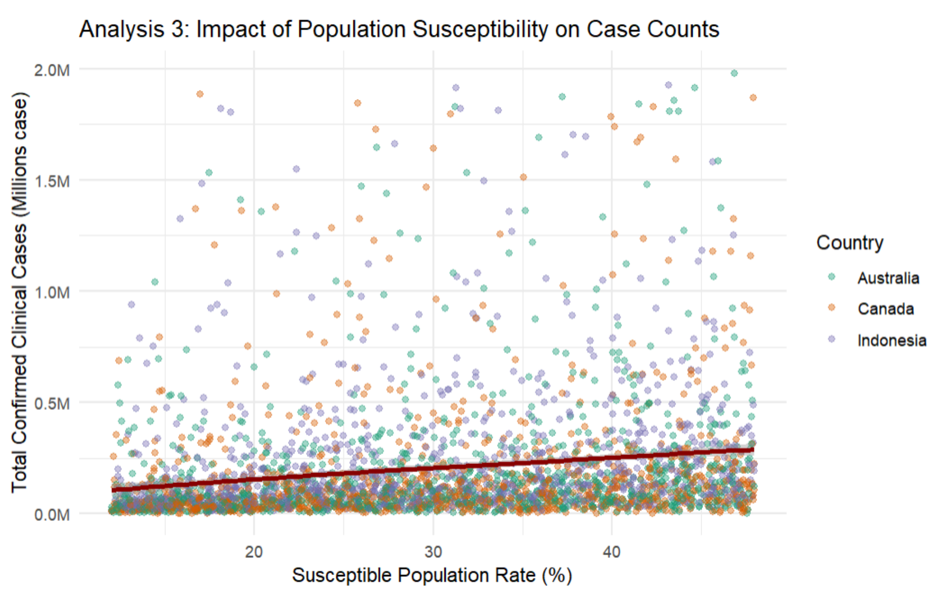
This graph proves that the models consider other environmental factors apart from those traditional ones that can contribute to the rise of the mosquito population, which alert the public to reduce stagnant water area, and awareness of mosquito population during high humidity weather.

B. Pesticide Control



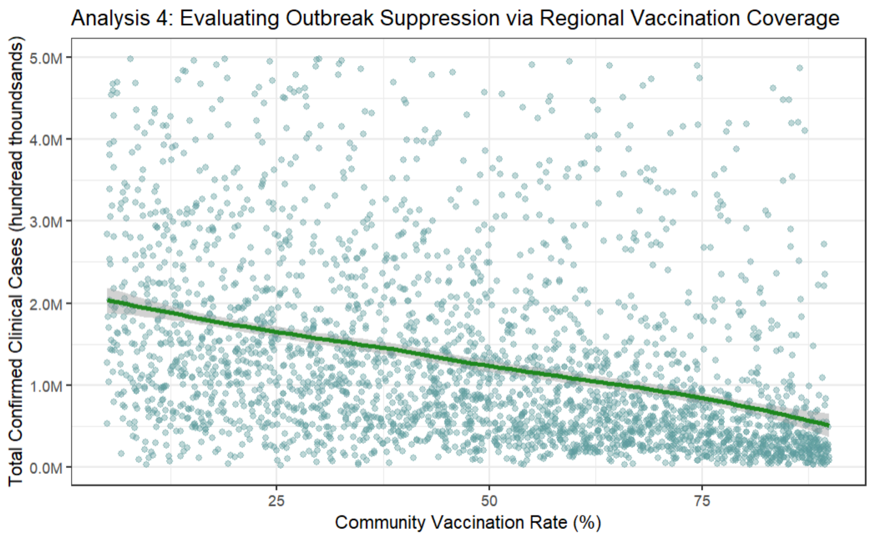
Visualization demonstrates that monthly pesticide sprays successfully suppress mosquito density spikes despite seasonal ecological drivers, validating the model's capacity to capture government interventions.

C. Population Susceptibility



A strong positive correlation confirms that higher local demographic susceptibility directly inflates clinical case counts, raising local awareness for proactive shielding.

D. Vaccination Rate



The plotted negative correlation identifies an immunization safety threshold, signaling a flag to warn of potential outbreaks in regions with low vaccination coverage.

E. Hospital Bed Infrastructure

A tibble: 9 × 5

Country <chr>	Province <chr>	Average_cases <dbl>	Average_Bed_Capacity <dbl>	Bed_Deficit_Ratio <dbl>
Indonesia	Jakarta	440360.3	6126.856	71.86206
Indonesia	West Java	390210.2	5914.786	65.96084
Canada	Quebec	391126.6	6147.669	63.61158
Canada	British Columbia	334738.7	5752.031	58.18477
Australia	Victoria	354029.6	6187.797	57.20492
Indonesia	Bali	341260.1	6175.678	55.24978
Australia	Queensland	329738.2	6269.567	52.58507
Australia	New South Wales	322080.3	6153.350	52.33376
Canada	Ontario	292944.8	6088.022	48.11032

9 rows

This table demonstrates the top 5 most vulnerable provinces that do not have enough facilities to handle all mosquito-borne disease. The “Bed_Deficit_Ratio” is the number that draws attention to local authorities to strategically expand hospital bed capacity.

5. Standard for Data Science Process, Data Governance and Management

5.1. Data Science Process

To ensure reliability, this project adheres to the 8-stage DataOne lifecycle model **[6]**.

Image 4: DataOne model **[6]**

Plan: Define forecasting input requirements, identify local data streams, and establish data-sharing agreements among public health and meteorological stakeholders.

Collect: Gather data from local meteorological stations, satellites, monthly hospital reports, and regional laboratory mosquito surveillance logs.

Assure: Implement strict quality checks to intercept anomalous data caused by weather sensor dropouts or human errors before downstream processing.

Describe: Document the mathematical logic and definitions of all incoming data properties, including types, units, and purposes.

Preserve: Securely back up and archive all historical data in cloud storage environments like AWS.

Discover: Enable stakeholders to easily search and retrieve specific datasets utilizing comprehensive metadata tags.

Integrate: Deploy ETL pipelines to merge diverse data formats categorized by year, month, country, and province.

Analyze: Feed integrated data into Apache Spark and Python-based multi-target machine learning models to forecast potential regional disease outbreaks.

5.2. Data Governance and Management

Data accessibility must be strategically tiered. While high-level disease risk maps and alert dashboards remain publicly accessible, raw hospital reports and laboratory data will be strictly restricted to authorized health professionals, data scientists, and local government officials. To guarantee instant data retrievability, comprehensive metadata tags will be generated based on time, region, and virus strain. For robust security, the project enforces end-to-end encryption and Role-Based Access Control (RBAC). All information is firmly encrypted during both data collection and cloud storage phases. RBAC restricts users to relevant data scopes; for instance, clinical medical staff cannot access raw meteorological reports but can still view high-level environmental summaries. During the validation process, the system ensures all incoming data is appropriate and mathematically logical by automatically detecting and eliminating missing values or incorrect data types. Regarding archiving, historical records exceeding 20 years will be compressed and secured in long-term storage, while remaining easily retrievable if required for longitudinal studies. To address ethical concerns, the careless handling of medical records containing personally identifiable information such as medical ID, addresses, name, severe privacy and re-identification risks. Therefore, any unnecessary personal information will be excluded during data collection and immediately removed out of the storage. Finally, to guarantee smooth project operations, the team will establish formal administrative protocols defining collaboration styles, individual role responsibilities, system maintenance pipelines, and crisis management strategies to ensure continuous operational alignment.

Reference

[1] M Khan, MT Mehran, ZU Haq, et al (2021). *Applications of artificial intelligence in COVID-19 pandemic: a comprehensive review, Expert Syst Appl, 185 (2021),* Article 115695. https://doi.org/10.1016/j.eswa.2021.115695

[2] WHO. (2025, December). *Malaria*. https://www.who.int/news-room/fact-sheets/detail/malaria#:~:text=Disease%20burden,malaria%20deaths%20in%20the%20Region.

[3] D-MOSS Dengue forecasting MOdel Satellite-based System. (n.d). https://www.hrwallingford.com/sites/default/files/inline-files/D-MOSS_factsheet_overview.pdf

[4] Department of Health. (2025, November). *Mosquito and mosquito-borne disease surveillance report*. https://www.health.vic.gov.au/infectious-diseases/mosquito-surveillance-report

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[6] DataONE.(n.d). *Data management best practices.*https://dataoneorg.github.io/Education/bestpractices/